

Tasks week1

Task1

Create a program named `main.py`

Create a program which calculate the area of a circle

```
from math import pi

# Press the green button in the gutter to run the script.
if __name__ == '__main__':
    r = float(input("Input the radius of the circle : "))
    print("The area of the circle with radius " + str(r) + " is: " + str(pi * r ** 2))
```

Task2 Simple Variable Declaration and Output

Create a program that stores an integer, a floating-point number, a string, and a boolean variable, and then outputs them.

Declare and initialize an integer variable `zahl` with the value 10. Check the variable's type afterward.

Declare and initialize a floating-point variable `kommazahl` with the value 10.5. Check the variable's type afterward.

Declare and initialize a string variable `text` with the value "Hello, World!". Check the variable's type afterward.

Declare and initialize a boolean variable `wahrheitswert` with the value True. Check the variable's type afterward.

Output all variables to the console with appropriate descriptions. Check the variable's type afterward

Task3 - Data Type Conversion

Create a program which will calculate the factorial

Create a program that converts different data types and outputs the results.

1. Convert an integer to a floating-point number.
2. Convert a floating-point number to an integer.
3. Convert an integer to a string.
4. Convert a string containing a number to an integer.
5. Convert an integer to a Boolean

Task4

Create a program that will calculate the factorial